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MEDICAL LECTURES

Fracture Healing

ادعوا لأخيك بالتوفيق، ولكم بالمثل

Bone Basics

- **Cortical (Compact) Bone:** Extremely dense structure. Provides core structural mechanical strength; undergoes remarkably **slow** biological **turnover**.
- **Cancellous (Trabecular) Bone:** Metaphyseal architecture. High surface area with **rapid** cellular **turnover** and remodeling rates.
- **Periosteum:** Outer fibrous protector + highly active inner cambium layer (dense with **osteoprogenitor cells & blood supply**). **Essential for primary calluses**.
- **Endosteum:** Lines the extensive medullary cavity; abundant source of active **osteoprogenitor** populations.
- **Blood Supply:** High reliance on the **nutrient artery**, **metaphyseal vessels**, and **periosteal vessels**. Disruption directly dictates necrosis and potential union failure.
- **Bone is uniquely regenerative:** it heals without scarring, but the choice of biological pathway is completely dictated by its mechanical environment (Stability).

1. Primary (Direct) Bone Healing

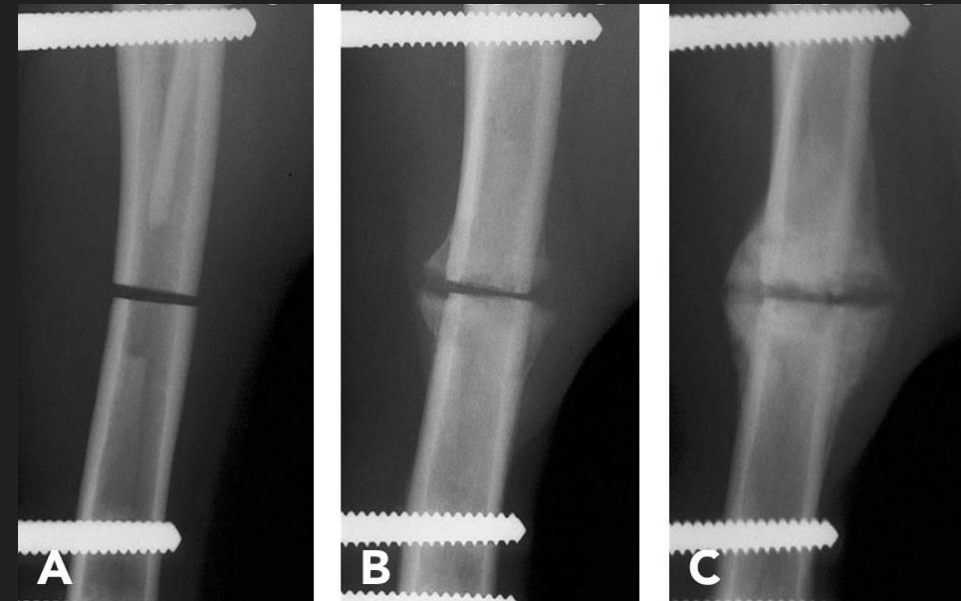
Occurs under conditions of absolute mechanical stability with compressed interfaces.

- **Stability:** **Absolute stability** (no motion allowed).
- **Mechanism:** Direct osteonal remodeling via **cutting cones**. Primary bone healing uses **Haversian remodeling** as its main mechanism.
 - **The Mechanism:** **Osteoclasts** group together to drill microscopic "**cutting cones**" directly across tightly compressed fracture lines, immediately followed by trailing **osteoblasts laying down strong lamellar bone**.
 - **Contact Healing:** Fracture gaps < 0.01 mm with zero motion. Cutting cones **directly restore** standard longitudinally oriented **Haversian systems**.
 - **Gap Healing:** Fracture gaps between 0.01 mm and 0.5 mm. **Woven bone** quickly **plugs the gap first**, then is gradually remodeled by **cutting cones** into structured lamellar bone.
- **Radiology:** **Zero visible external callus**.
- **Surgical Example:** **Compression plating**, rigid lag screws.
- **Clinical Warning:**
 - **Extremely slow** healing path.
 - **No warning callus on X-ray**.
 - **Complete** biomechanical **reliance on the surgical hardware**.

2. Secondary (Indirect) Bone Healing

The standard evolutionary pathway triggered under conditions of relative mechanical stability.

- **Stability:** **Relative stability** (controlled micro-motion).
- **Mechanism:** **Callus formation** followed by **ossification**.
 - The bone handles local mechanical movement by building a **progressively stiffer tissue bridge**.
- **Radiology:** Large, robust, **visible callus on standard X-ray**.
 - Radiological Progression: Moves predictably from fluffy cartilage shadow (Soft Callus) to mineralized woven bone (Hard Callus) and final restoration.
- **Surgical Example:** **Intramedullary nails**, **cast immobilization**, bridge plates.
- ✓ This is the default biological path in natural states and when standard casts, external fixators, or intramedullary nails are applied.



Stage 1: Hematoma & Inflammation



Timeframe: **Hours to Days**. Ruptured local vessels form clot; local sterile necrotic **inflammatory cascade recruits stem cells**.

1. **Vascular Emergency:** Severed intramedullary and periosteal blood vessels result in **immediate localized hematoma**. Hypoxia leads to death of bone ends.
2. **Chemical Cascade:** Degranulating platelets release highly potent growth factors like **PDGF** and **TGF- β** .
3. **Cellular Recruitment:** Influx of **neutrophils** followed immediately by critical **macrophages**.
4. **Granulation Bed:** Macrophages clear necrotic bone debris and release **angiogenic/osteogenic signals** (VEGF, BMPs, IL-6), creating initial highly cellular granulation tissue.

Stage 2: Soft Callus Formation



Timeframe: Days to Weeks. Mesenchymal cells differentiate into chondrocytes, laying down cartilage bridge to lower strain.

1. **Progenitor Recruitment:** Undifferentiated mesenchymal stem cells from the active cambium periosteum and deep endosteum migrate into the injury bed.
2. **Chondrogenesis:** Under mechanical motion and relative hypoxia, progenitor cells express transcription factor SOX9, differentiating into chondrocytes.
3. **Cartilage Anlage:** Chondrocytes secrete a flexible cartilaginous matrix (soft callus) that physically bridges the fracture gap.
4. **Peripheral Intramembranous Ossification:** Farther away from active movement, where strain is lowest, progenitor cells directly differentiate into osteoblasts, depositing osteoid.

Stage 3: Hard Callus Formation



Timeframe: Weeks to Months. Endochondral ossification replaces weak cartilage with solid, disorganized woven bone.

1. **Cartilage Remodeling:** Chondrocytes within the soft callus undergo marked hypertrophy, release alkaline phosphatase, and calcify their surrounding matrix (soft callus).
2. **Vascular Invasion:** Hypoxic signaling triggers robust VEGF secretion. An invading vascular capillary sprout front breaches the calcified matrix.
3. **Osteogenesis:** Sprouted vessels carry osteoblasts that use the calcified cartilage fragments as scaffolds to lay down structurally disorganized woven bone.
4. **Clinical Union:** Woven bone forms a solid, non-mobile skeletal bridge. Movement under load ceases completely.

Stage 4: Structural Bone Remodeling

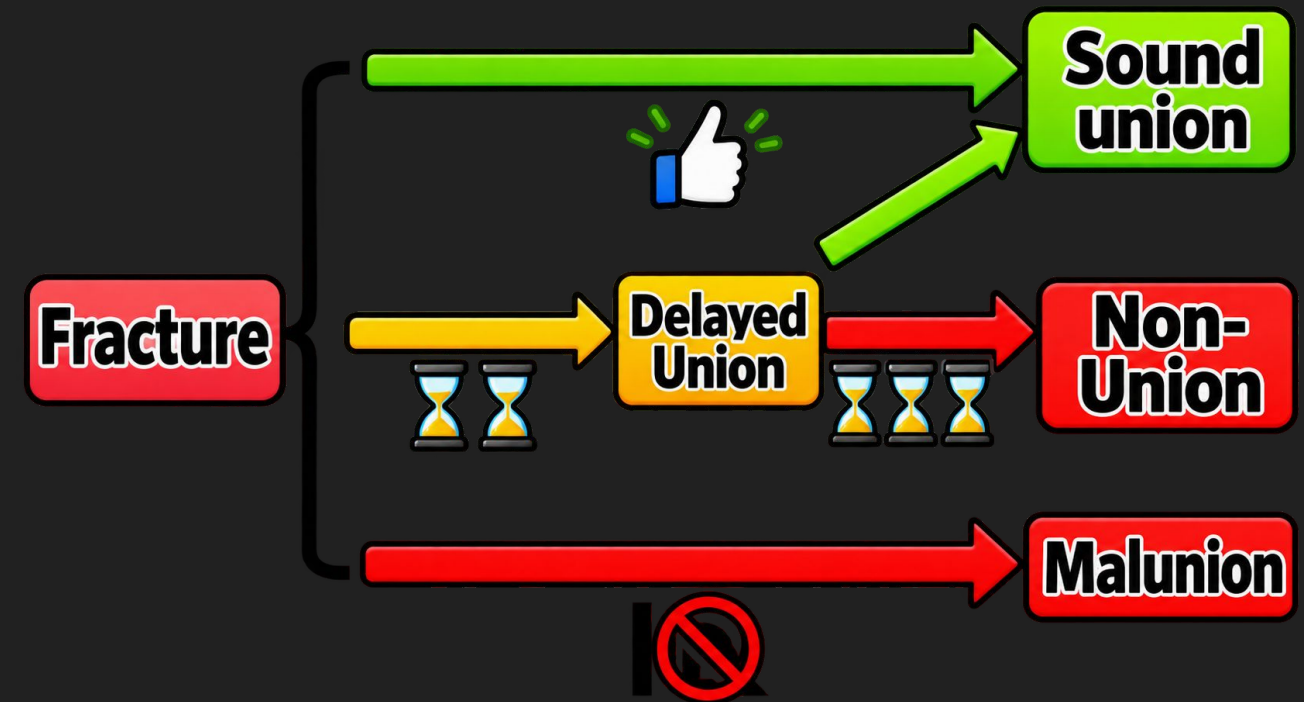


Timeframe: Months to Years. Wolff's Law takes over. Osteocytes reshape the structure, restoring the original canal.

- **Lamellar Transition:** Highly active osteoclasts and osteoblasts slowly replace weak woven bone with structured lamellar bone.
- **Mechanical Stress Alignment:** Newly formed trabeculae are laid down specifically along the axes of physiological weight-bearing.
- **Reconstruction:** Resorption of excess external callus restores the bone's original contour and medullary canal.

3. Union

- **Clinical Evaluation of Union:**
 - Absence of abnormal movement at the fracture site.
 - Absence of local tenderness on direct palpation.
 - Complete ability to bear physiological weight on the affected limb without severe pain.
- **Radiographic Indicators:**
- Visualization of bridging bone across at least 3 out of 4 cortices on high-quality orthogonal AP and Lateral X-rays.
- Gradual blurring and complete disappearance of the fracture line.



4. Complications of Fracture Healing

- Most fractures heal well, but unfortunately some of them experience various complications at the healing phase.
- Those complications may be related to the fracture cause and pattern, patient's condition, or miss management.
- Prevention, early diagnosis and adequate management of these complications are of most importance for the patient to have a satisfying outcome.

I. Delayed Union

- Delayed union occurs when **a bone fails to unite within an average anticipated time**. This varies with location and configuration as well as the specific bone and age group.

I. Causes of Delayed Union

A. Biological causes

- Inadequate blood supply.
- Severe soft tissue damage.
- Infection.

B. Biomechanical Causes

- Fracture gap.
- Unstable fixation.
- Over-rigid fixation.
- Soft tissue interposition.

C. Patient-related Causes

- Age.
- Smoking.
- Alcohol use.
- Poor nutrition.
- Debilitating diseases.
- Medications (corticosteroids, NSAIDs and cytotoxic immunosuppressants).

★ II. Clinical Picture of Delayed Union

- Fracture **pain and tenderness** persist.
- Fracture line** remains **visible** on X-ray.
- Bone ends are **not sclerosed** or **atrophic**.

III. Treatment of Delayed Union → Eliminate the Cause

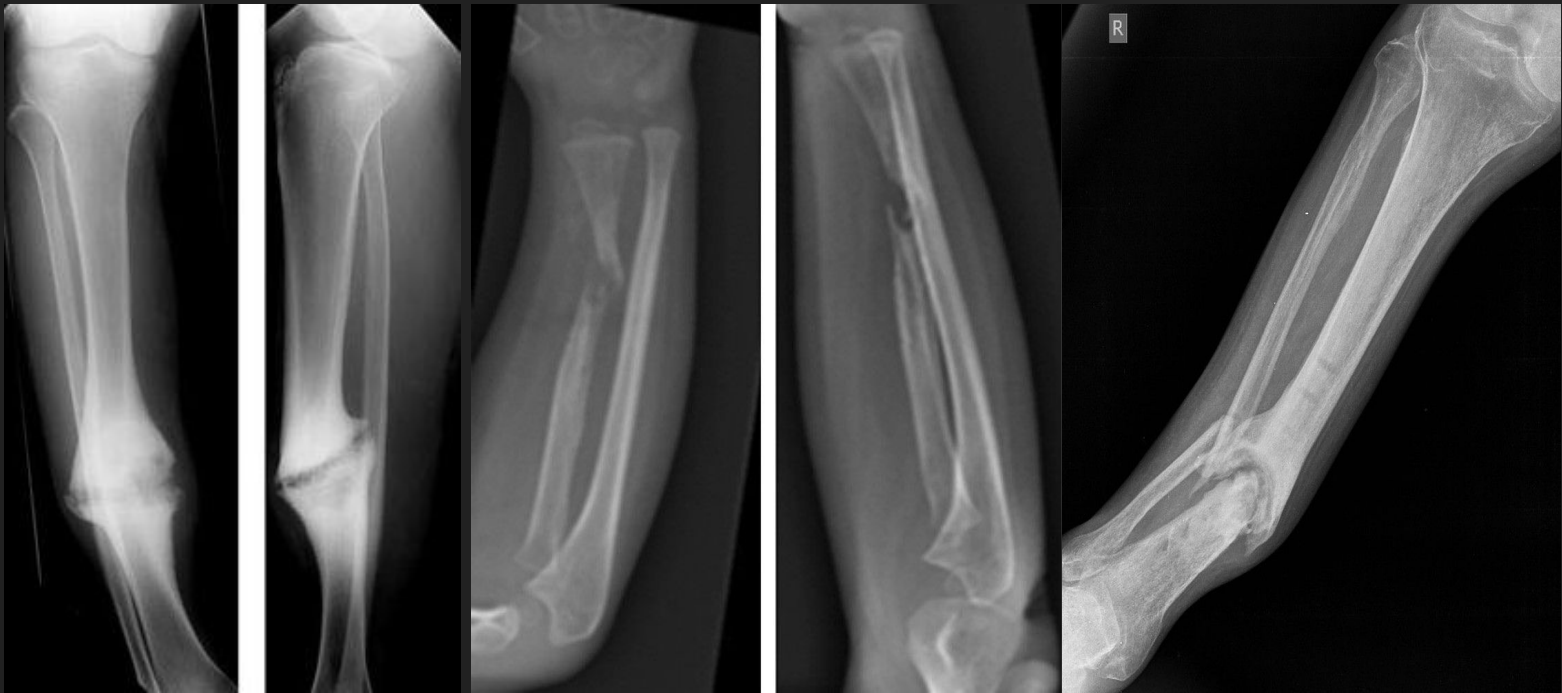
- Biological:** Bone graft, Soft tissue reconstruction, Treat infection.
- Biomechanical:** Bone graft, Adequate fixation, Surgically remove any soft tissue interposition.
- Patient-related:** Improve general condition, Control debilitating diseases, Stop smoking and alcohol, Stop NSAIDs.

II. Non-union

- Some cases of delayed union gradually turns into non-union.
- FDA defines it as a fracture that is **at least 9 months** old and **has not shown any signs of healing for 3 consecutive months**.
- Same causes of delayed union.
- It is either Aseptic Non-Union or Septic Non-Union.

≡ Types of Aseptic Non-Union ≡

Type	Features	Management
Hypertrophic 	Widened bone ends by hypertrophic callus due to unstable fracture fixation.	Provide appropriate stabilization via internal fixation.
Atrophic 	- Avascular or hypovascular bone ends with absent callus formation (biological failure). - Fracture ends are atrophic.	Bone graft (autologous preferred). Stable fixation. - Creation of a healthy well-vascularized local soft-tissue envelope.
Pseudarthrosis 	- Formation of a “false joint”. - Bone ends are atrophic with impaired vasculature.	Complete surgical takedown with excision of the atrophic bone ends, followed by proper surgical stabilization.



Hypertrophic

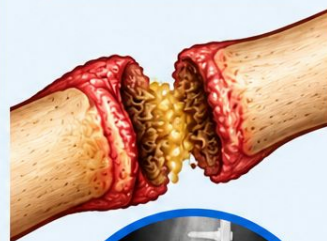
Atrophic

Pseudarthrosis

Septic Non-Union

Type

Septic Non-Union



Features

- **Clinical signs of infection** (tenderness, erythema, warmth, swelling).
- **Inflammatory markers:** CBC, ESR, and CRP.

Management

- **Debridement:** removal of all infected, devitalized bone, soft tissue, and implants until healthy, bleeding bone ends and vascularized bed are reached.
- **Stabilization:** Stable fixation via external fixation.
- **Reconstruction:** Bone transport or later limb lengthening via the Ilizarov technique (no graft).
- **Antibiotics:** According to culture.



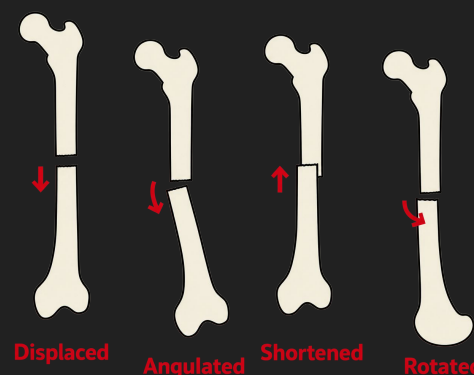
Septic Non-Union

III. Malunion

- Bone fragments heal in an **unsatisfactory position**.
- Can be in the form of angulation, rotation or shortening.
- The **range of accepted deformity varies** according to the type of deformity, patient's age, fractured bone, place of the fracture within the bone (e.g. intra-articular, metaphyseal, diaphyseal).

I. Causes of Malunion

- **Negligence** or **misdiagnosis**.
 - **Failure of proper reduction**.
 - **Failure to hold reduction during healing process** (after operative or nonoperative fixation).
- ✓ It is a must to do X-rays to check the position of the fracture while it is uniting, especially during the first 2-3 weeks when the situation may change without warning.



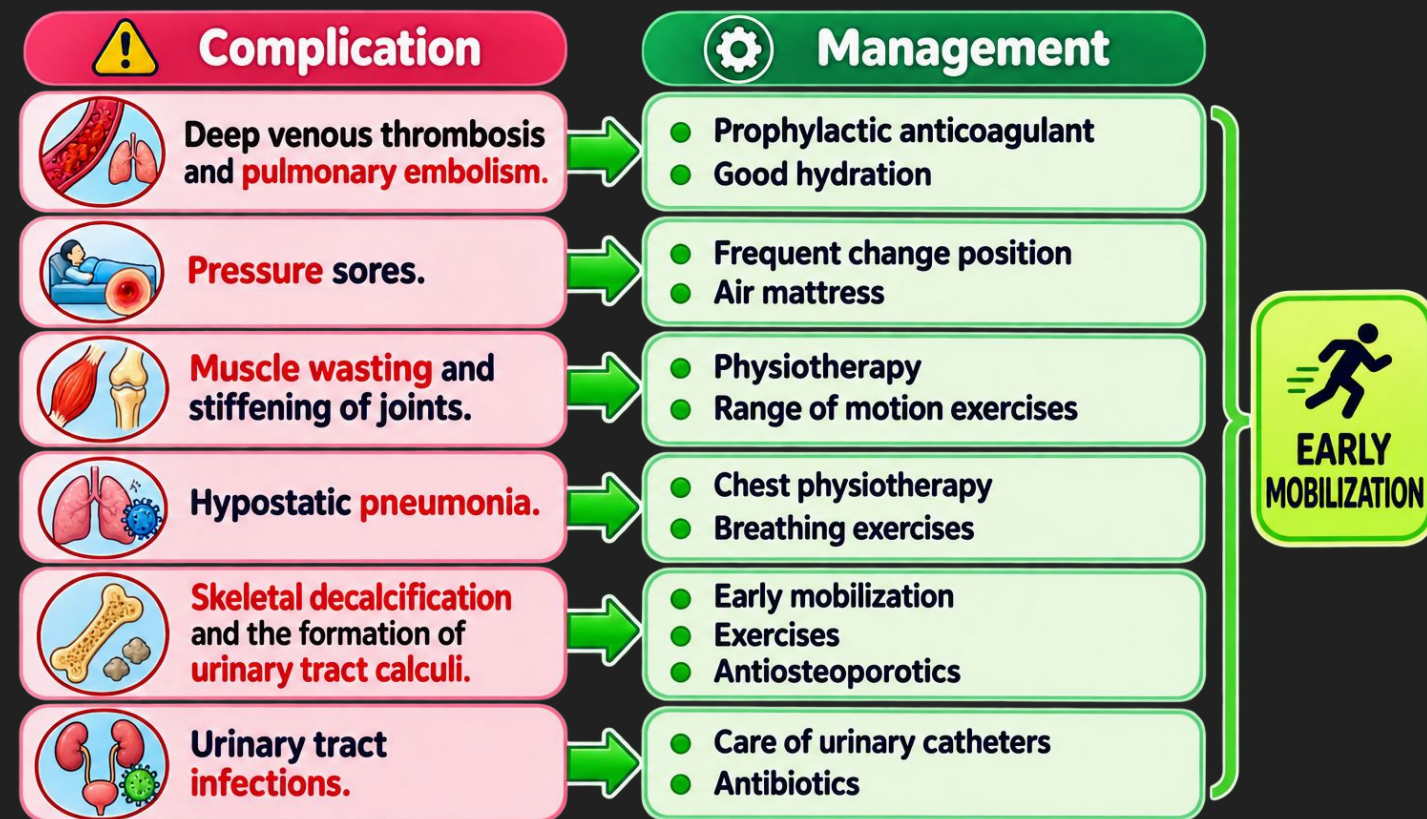
I. Management of Malunion

- Prevention by **close follow-up to early detect any unacceptable displacement**.
- Malunion with unacceptable displacement (according to age, type and amount of fracture displacement) should be corrected surgically by **osteotomy, reduction and proper fixation**.

IV. Complications of Prolonged Recumbency

- After spine, pelvis or lower limb fractures patients may be unable to walk for variable amounts of time.
- **All efforts should be taken to shorten this period to avoid complications of prolonged recumbency.**
- **Early ambulation** as tolerated should be a **primary goal** in the management plan of any trauma patient.

Complications of Prolonged Recumbency



V. Joint Complications



Joint Complications

1

Joint Stiffness



Causes

- Prolonged joint immobilization which happens due to:
- Hemarthrosis.
- Periarticular structures edema.
- Soft tissue adhesions.
- Improper position of joint fixation.



Prevention

- ✓ Keep the joints **mobile** as possible.
- ✓ Mind '**position of safe immobilization**'.

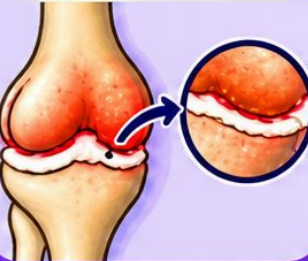


Management

- ✓ If stiff → prolonged physiotherapy → arthroscopic-guided releases → operation.

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Joint Osteoarthritis



Causes

- Cartilage damage.
- Articular surface step.
- Malunited metaphyseal fracture.



Prevention

- ✓ Intra-articular fractures should be **reduced** and **fixed** properly.
- ✓ Joint alignment should be **maintained, monitored** and if needed **restored** during fracture healing.



Management

- ✓ Management of **osteoarthritis** if it happens.